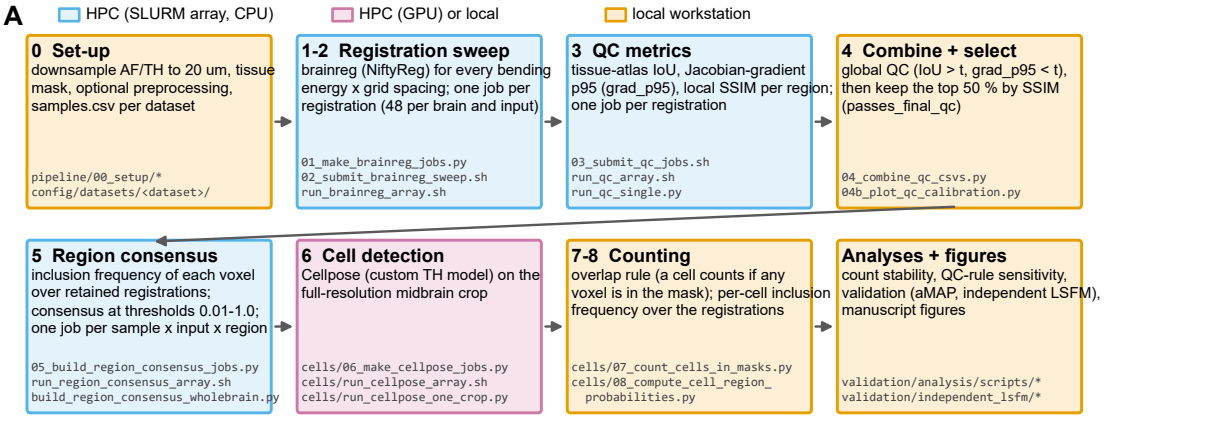
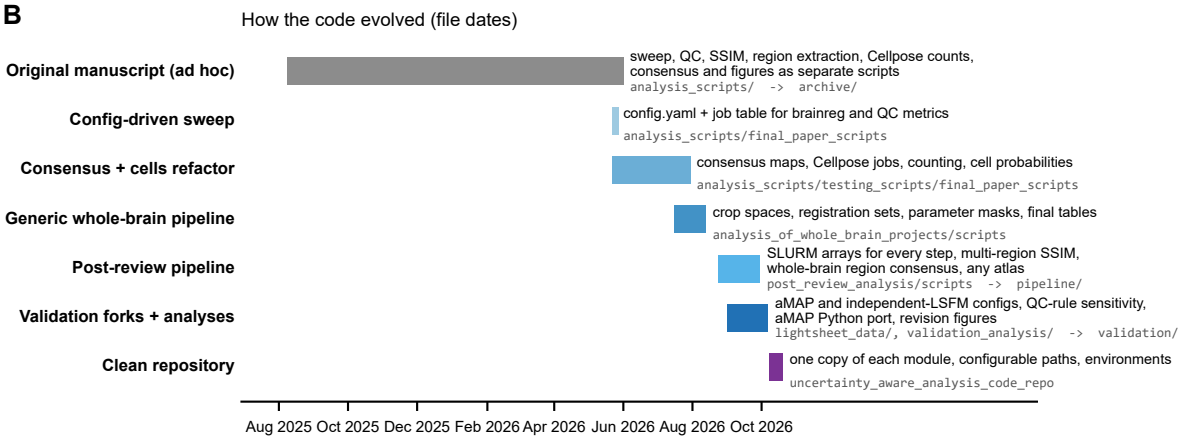
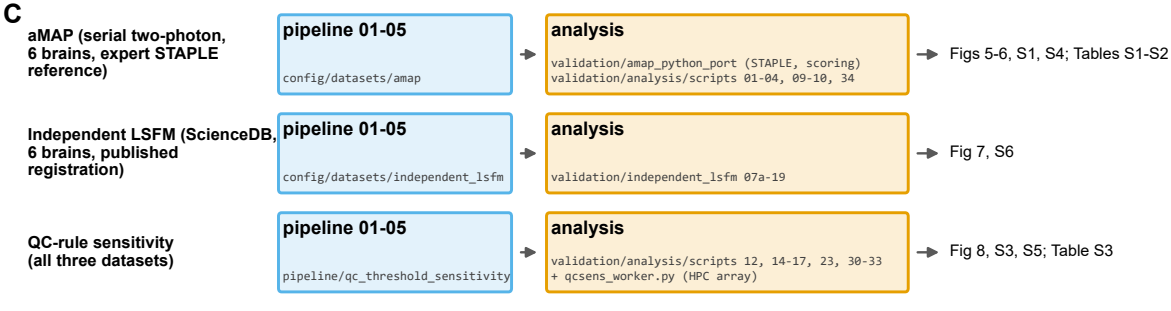

A
HPC (SLURM array, CPU)
HPC (GPU) or local
local workstation
0 Set-up
downsample AF/TH to 20 um, tissue mask, optional preprocessing, samples.csv per dataset
pipeline/00_setup/*
config/datasets/<dataset>/
1-2 Registration sweep
brainreg (NiftyReg) for every bending energy x grid spacing; one job per registration (48 per brain and input)
01_make_brainreg_jobs.py
02_submit_brainreg_sweep.sh
run_brainreg_array.sh
3 QC metrics
tissue-atlas IoU, Jacobian-gradient p95 (grad_p95), local SSIM per region; one job per registration
03_submit_qc_jobs.sh
run_qc_array.sh
run_qc_single.py
4 Combine + select
global QC (IoU > t, grad_p95 < t), then keep the top 50 % by SSIM (passes_final_qc)
04_combine_qc_csvs.py
04b_plot_qc_calibration.py
5 Region consensus
inclusion frequency of each voxel over retained registrations; consensus at thresholds 0.01-1.0; one job per sample x input x region
05_build_region_consensus_jobs.py
run_region_consensus_array.sh
build_region_consensus_wholebrain.py
6 Cell detection
Cellpose (custom TH model) on the full-resolution midbrain crop
cells/06_make_cellpose_jobs.py
cells/run_cellpose_array.sh
cells/run_cellpose_one_crop.py
7-8 Counting
overlap rule (a cell counts if any voxel is in the mask); per-cell inclusion frequency over the registrations
cells/07_count_cells_in_masks.py
cells/08_compute_cell_region_probabilities.py
Analyses + figures
count stability, QC-rule sensitivity, validation (aMAP, independent LSFM), manuscript figures
validation/analysis/scripts/*
validation/independent_lsfm/*
HPC SLURM arrays: 234 logged registrations took 92 h of compute but 3.0 h of wall time (31x)


B
How the code evolved (file dates)
Original manuscript (ad hoc)
sweep, QC, SSIM, region extraction, Cellpose counts, consensus and figures as separate scripts
analysis_scripts/ -> archive/
Config-driven sweep
config.yaml + job table for brainreg and QC metrics
analysis_scripts/final_paper_scripts
Consensus + cells refactor
consensus maps, Cellpose jobs, counting, cell probabilities
analysis_scripts/testing_scripts/final_paper_scripts
Generic whole-brain pipeline
crop spaces, registration sets, parameter masks, final tables
analysis_of_whole_brain_projects/scripts
Post-review pipeline
SLURM arrays for every step, multi-region SSIM, whole-brain region consensus, any atlas
post_review_analysis/scripts -> pipeline/
Validation forks + analyses
aMAP and independent-LSFM configs, QC-rule sensitivity, aMAP Python port, revision figures
lightsheet_data/, validation_analysis/ -> validation/
Clean repository
one copy of each module, configurable paths, environments
uncertainty_aware_analysis_code_repo
Aug 2025 Oct 2025 Dec 2025 Feb 2026 Apr 2026 Jun 2026 Aug 2026 Oct 2026


C
aMAP (serial two-photon, 6 brains, expert STAPLE reference)
pipeline 01-05
config/datasets/amap
analysis
validation/amap_python_port (STAPLE, scoring)
validation/analysis/scripts 01-04, 09-10, 34
Figs 5-6, S1, S4; Tables S1-S2
Independent LSFM (ScienceDB, 6 brains, published registration)
pipeline 01-05
config/datasets/independent_lsfm
analysis
validation/independent_lsfm 07a-19
Fig 7, S6
QC-rule sensitivity (all three datasets)
pipeline 01-05
pipeline/qc_threshold_sensitivity
analysis
validation/analysis/scripts 12, 14-17, 23, 30-33
+ qcsens_worker.py (HPC array)
Fig 8, S3, S5; Table S3